Customer
Front End
LoadBalancer
API Gateway
User Service
Product Service
Order Service
PostgreSQL

alt
[good credentials]
Customer wants to login(email, password)
route Request()
POST /login (email, password)
POST /login (email, password)
get User (email, password)
get User(): User
user Found (): Person
get User(): User
200 - OK
Display login confirmation, User is signed in()
distribute Request()
200 - OK

[wrong credentials]
User logs with wrong credentials (Wrong email / password)
route Request()
POST /login (wrong email, password)
POST /login (wrong email, password)
get User (wrong email, password)
User Not Found Exception(): 401 - Unauthorized
not Found()
User Not Found Exception(): 401 Unauthorized
Wrong credentials, try again()
distribute Request()
401 - Unauthorized

loop
[customer lists through products until he chooses]
Customer wants to list products (fruits)
route Request()
Validation()
GET /products (fruits)
/products (fruits)
get Products (fruits)
Products()
products()
List all products for given parameters()
distribute Request()
200 - OK
products Array()
200 - OK

Wants to buy product (Order Data)
route Request()
Validation()
POST /products/{id}/orders (Orderdata)
POST /products/{id}/orders (Orderdata)
alt
[product is in stock]
Valid Data Product()
Product Stock (stock)
ProductIsIn Stock()
Save Order (Order)
Order saved()
Order Summary()
[product is not in stock]
Valid Data Product()
Product Stock (stock)
Product Not In Stock (Exception)
Order Summary()
200 - OK
Displays confirmation and summary for eshop order()
distribute Request()
200 - OK

route Request()
ProductIsNotIn Stock choose other one()
distribute Request()
400 - Bad Request
ProductNotInStockException(): 400 - Bad Request
ProductNotIn StockException()